Limitations and Ethics Assessment

Main limitations:
The model is trained on only 50 users, so performance can still vary materially under a different split or a larger sample.
The demographic table is only partially real: 20 users have provided forms and 30 users have generated placeholders.
The target itself is derived from TX_Per_Month quantiles, so the model is learning a constructed segmentation rather than an external ground-truth label.

Ethics and bias considerations:
Demographic variables may encode socioeconomic patterns that could create unfair targeting if used without safeguards.
Mobile-money behavior may reflect access constraints, geography, and device ownership, not just user preference or value.
The data is anonymized, but re-identification risk still exists if external information is combined with detailed transaction behavior.

How the model should be used:
Use it for exploratory segmentation, portfolio analysis, and product planning.

How the model should not be used:
Do not use it alone for punitive decisions, exclusion, credit denial, or any high-stakes individual judgment.

Future work:
Collect real demographics for all cohorts, enlarge the participant base, test repeated cross-validation, and validate the segment definitions with domain stakeholders.